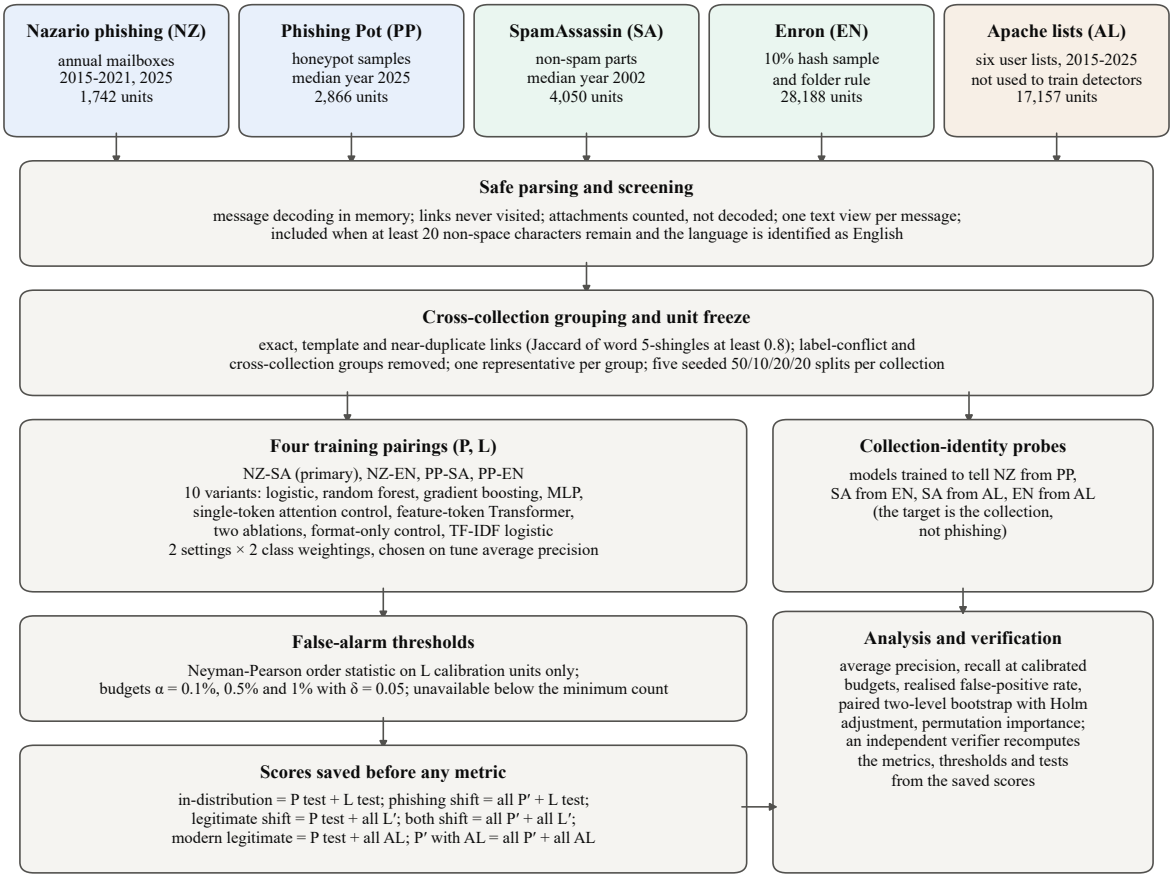

Nazario phishing (NZ)
annual mailboxes
2015-2021, 2025
1,742 units
Phishing Pot (PP)
honeypot samples
median year 2025
2,866 units
SpamAssassin (SA)
non-spam parts
median year 2002
4,050 units
Enron (EN)
10% hash sample
and folder rule
28,188 units
Apache lists (AL)
six user lists, 2015-2025
not used to train detectors
17,157 units
Safe parsing and screening
message decoding in memory; links never visited; attachments counted, not decoded; one text view per message;
included when at least 20 non-space characters remain and the language is identified as English
Cross-collection grouping and unit freeze
exact, template and near-duplicate links (Jaccard of word 5-shingles at least 0.8); label-conflict and
cross-collection groups removed; one representative per group; five seeded 50/10/20/20 splits per collection
Four training pairings (P, L)
NZ-SA (primary), NZ-EN, PP-SA, PP-EN
10 variants: logistic, random forest, gradient boosting, MLP,
single-token attention control, feature-token Transformer,
two ablations, format-only control, TF-IDF logistic
2 settings × 2 class weightings, chosen on tune average precision
Collection-identity probes
models trained to tell NZ from PP,
SA from EN, SA from AL, EN from AL
(the target is the collection,
not phishing)
False-alarm thresholds
Neyman-Pearson order statistic on L calibration units only;
budgets α = 0.1%, 0.5% and 1% with δ = 0.05; unavailable below the minimum count
Scores saved before any metric
in-distribution = P test + L test; phishing shift = all P′ + L test;
legitimate shift = P test + all L′; both shift = all P′ + all L′;
modern legitimate = P test + all AL; P′ with AL = all P′ + all AL
Analysis and verification
average precision, recall at calibrated
budgets, realised false-positive rate,
paired two-level bootstrap with Holm
adjustment, permutation importance;
an independent verifier recomputes
the metrics, thresholds and tests
from the saved scores